

FantasyXI

An Intelligent Decision-Support Framework for Fantasy Premier League

Google Summer of Code 2026

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Project Size	350 Hours (Large Project)

About Me

I am an undergraduate student passionate about combining machine learning with competitive, data-rich domains. Fantasy Premier League is exactly that: millions of managers making sequential decisions under uncertainty, budget constraints, and direct competition with each other. I am drawn to FantasyXI because it lets me apply deep learning, game theory, and reinforcement learning together to a single, well-defined competitive system, rather than treating each technique in isolation.

Relevant Background

- Strong foundation in Data Structures, Algorithms, Probability & Statistics, Artificial Intelligence, and Database Management Systems.
- Hands-on machine learning project experience, including time-series and sequence models.
- Active open-source contributor with a track record of shipping complete, documented pipelines.
- Comfortable with PyTorch, scikit-learn, and optimisation libraries (PuLP) used throughout this proposal.

Availability

I can commit 25–30 hours per week for the duration of the program, with the ability to extend during my summer vacation window. I will provide biweekly blog updates, attend all mentor check-ins, and flag any reduced-availability periods (e.g. examinations) well in advance so milestones stay on track.

1. Project Description

1.1 About

Fantasy Premier League (FPL) is one of the world's most popular fantasy sports platforms, where managers continuously optimise squad selection, transfers, and captaincy under a fixed budget and transfer-cost rules, while competing against millions of other managers. Existing tools largely rely on short-horizon historical statistics and do not explicitly model the long-term, sequential, and competitive nature of the game. FantasyXI proposes a unified decision-support framework that combines **deep learning** for performance forecasting, **game theory** for ownership-aware, rank-sensitive recommendations, and **reinforcement learning** for sequential transfer and captaincy planning across a full season.

1.2 Objectives

- **O1.** Build a reliable historical + live data pipeline from the official FPL API covering player statistics, fixtures, ownership, and price changes.
- **O2.** Train a BiLSTM/Attention deep learning model that forecasts each player's expected points over the next 1–5 gameweeks from time-series performance features.
- **O3.** Develop a game-theoretic scoring layer that converts raw point forecasts and ownership data into rank-aware recommendations — distinguishing 'template' (rank-protecting) from 'differential' (rank-gaining) picks.
- **O4.** Formulate transfers and captaincy as a Markov Decision Process and train a reinforcement learning agent that optimises cumulative season performance rather than greedy single-gameweek gains.
- **O5.** Combine all three components into a single, reproducible, open-source FPL management assistant validated against real historical seasons.

1.3 Deliverables

ID	Deliverable
D1	Automated FPL data pipeline producing clean, versioned per-gameweek player time series.
D2	BiLSTM + Attention forecasting model predicting next-gameweek and multi-gameweek expected points, benchmarked against a baseline.
D3	Game-theoretic recommendation engine producing a ranked list of template vs. differential picks for any risk-tolerance parameter.
D4	RL transfer/captaincy agent (DQN) trained on simulated seasons, benchmarked against a greedy points-maximising baseline.
D5	ILP-based squad optimizer enforcing official FPL rules (budget, club limits, formation), used as the action filter for both RL and recommendation engines.
D6	Backtested validation report comparing FantasyXI's recommendations against actual outcomes from a completed historical season.

2. Implementation

2.1 Data Collection & Feature Engineering

Data is pulled from the official FPL public API (*bootstrap-static*, *fixtures*, and *element-summary* endpoints), producing per-player, per-gameweek records of minutes, goals, assists, Expected Goals (xG), Expected Assists (xA), Expected Goal Involvements (xGI), Expected Goals Conceded (xGC), ICT index, bonus points, ownership percentage, and price. Rolling 3- and 5-gameweek form windows, ownership momentum, and price-change features are engineered, and the series are reshaped into sliding windows for sequence modelling.

2.2 Deep Learning Forecaster — BiLSTM + Attention

A 2-layer Bidirectional LSTM encodes the last 5 gameweeks of each player's feature vector, with an additive attention layer over the hidden states so the model can weight the most informative recent gameweeks (e.g. a sudden injury return or a favourable fixture run) more heavily than older ones. A small dense head maps the attended context vector to a single expected-points prediction. The model is trained separately for short (1 GW) and medium (5 GW) horizons, and evaluated using R^2 and MAE per position, mirroring the evaluation style used in prior open FPL forecasting work such as OpenFPL.

2.3 Game Theory — Differential Ownership & Rank-Gain Scoring

Raw expected points alone do not capture what actually moves a manager's rank: a highly-owned premium player protects rank but rarely gains it, while a low-ownership differential can swing rank significantly if it outperforms. FantasyXI scores each player using an ownership-weighted expected rank-gain function:

$$\text{rank_gain_score}(p) = (\text{EV}(p) - \text{field_EV}) \times (1 - \text{ownership}(p)) - \lambda \times \sqrt{\text{variance}(p)} \times (1 - \text{ownership}(p))$$

where λ is a user-controlled risk-tolerance parameter: $\lambda \rightarrow 0$ favours rank-protecting template picks, while $\lambda \rightarrow 1$ favours high-upside differentials. This directly operationalises the findings of 'Differential Ownership and Fantasy Football Strategy' and 'Competing with Humans at Fantasy Football' into an actionable scoring layer.

2.4 Reinforcement Learning — Sequential Transfer & Captaincy Planning

Transfers and captaincy choices are modelled as a Markov Decision Process: the **state** captures the current squad's short-horizon forecasted EV, bank balance, free transfers, and chip availability; the **action** space covers hold / single-transfer / point-hit double-transfer decisions (filtered through the ILP squad optimizer so every action stays rule-compliant); the **reward** is realised gameweek points minus any transfer-hit penalty. A Deep Q-Network is trained on simulated seasons built from the forecaster's predicted EV, learning to trade short-term points for long-term squad value rather than greedily maximising every single gameweek.

2.5 Squad Optimization

An Integer Linear Program (solved with PuLP / CBC) enforces the official FPL constraints — 15-man squad (2 GK / 5 DEF / 5 MID / 3 FWD), $\leq £100\text{m}$ budget, maximum 3 players per real-world club, and a valid starting-XI formation — while maximising total expected points of the starting XI plus a doubled contribution from the optimal captain. This module is shared between the stand-alone squad-builder tool and the RL agent's action filter.

2.6 System Architecture

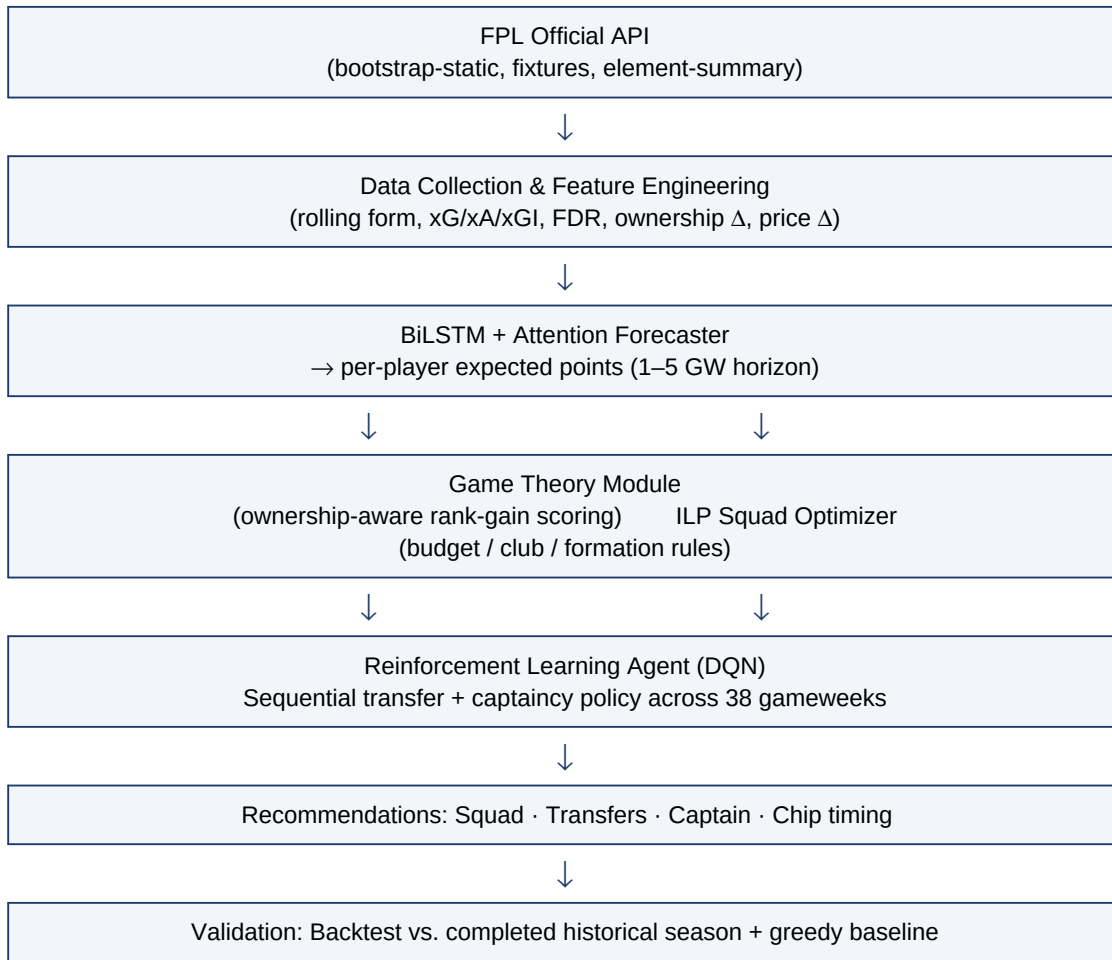


Figure 1: End-to-end FantasyXI pipeline — data flows top to bottom from raw API ingestion through forecasting, game-theoretic and rule-based filtering, into the RL policy that issues the final weekly recommendation.

3. Project Timeline (350 Hours)

Phase	Duration	Key Tasks	Milestone
Community Bonding	~2 weeks	Study FPL API & OpenFPL baseline; review prior FPL ML literature; set up repo and dev data/planning workflow	Discussions on report and dev data/planning workflow
Phase 1: Data Pipeline	3 weeks	Build FPL API collector; engineer rolling-form and fixture-difficult-to-handle edge cases; construct sliding window sequence	Build a robust, constructed sliding window sequence
Phase 2: Forecasting Models	2 weeks	Implement BiLSTM+Attention model; train 1-GW and 5-GW horizons; evaluate on 1000 random splits	20k Forecasts evaluated & 100k MAE per position MD
Phase 3: Game Theory Models	3 weeks	Implement ownership-aware rank-gain scoring; calibrate risk-to-loss; evaluate on 1000 random splits	10k Recommendations; backtests on 1000 splits. different
Phase 4: Squad Optimizer	2 weeks	Implement ILP squad/formation optimizer with PuLP; integrate with Rule-based optimizer; validate on aggregated	10k Rule-based optimizer; validate aggregated
Phase 5: RL Agent	4 weeks	Build season simulation environment; implement DQN agent for 1000 random splits	10k RL agent training; benchmark against sim
Phase 6: Backtesting & Validation	3 weeks	Run full pipeline on a completed historical FPL season; compare predicted squad cost/points to actual outcomes	10k Validation on 1000 splits; compare to actual outcomes
Phase 7: Final Report	~2 weeks	Clean and document all code; write final report and demo notebook; prepare presentation; address mentor feedback	Final report, presentation; address mentor feedback

4. Evaluation & Benchmarks

- **Forecasting:** R^2 and MAE per position and per horizon (1 GW vs 5 GW), benchmarked against the FPL API's own predicted-points field and the open-source OpenFPL method.
- **Game theory:** backtested rank-gain realised by template vs. differential recommendations across a completed historical season, at multiple risk-tolerance (λ) settings.
- **RL agent:** total simulated season points and transfer-hit efficiency vs. a greedy points-maximising baseline with no foresight, over a full 38-gameweek simulated season.
- **End-to-end:** final squad/transfer recommendations compared against actual high-rank human manager decisions for the same historical season.

5. My Motivation

Machine learning and football analytics are two of my long-standing interests, and FPL is a uniquely rich environment to combine them: it is a real, large-scale, competitive system with public data, a well-defined reward signal (points and rank), and genuinely hard sequential decision-making. Building FantasyXI gives me the opportunity to go beyond a single point-prediction model and build a complete, reproducible, open-source decision-support system — one that explicitly reasons about uncertainty, competition, and the long-term consequences of every transfer and captaincy choice.

6. References

- Fantasy Premier League Official Website & API — <https://fantasy.premierleague.com/>
- OpenFPL: An Open-Source Forecasting Method Rivaling State-of-the-Art Fantasy Premier League Services.
- Time Series Modeling for Dream Team in Fantasy Premier League.
- Differential Ownership and Fantasy Football Strategy.
- Competing with Humans at Fantasy Football: Team Formation and Management.